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DSC680 – Applied Data Science
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March 8, 2026

Q&A

1. How were screen-time variables harmonized across datasets that define use differently?
    a. Screen-time variables were mapped into one common hours-per-day schema: total, social, gaming, streaming, and education. If a source only supported total screen exposure, it was used for the total measure only. Outcomes were also aligned to consistent scales, and the mental-health index was built from standardized anxiety and depression scores.
2. Why was 7.5 hours selected as the poor-sleep threshold, and how sensitive are results to that cutoff?
    a. The 7.5-hour cutoff gave a practical and interpretable binary sleep-risk threshold for the pilot model. The result was stable to reasonable changes: at less than 7.5 hours the odds ratio per extra screen hour was 2.256, at less than 7.0 hours it was 2.164, and at less than 8.0 hours it was 1.914. The conclusion stayed the same across cutoffs.
3. Do associations differ by age group, sex, or socioeconomic status after adjustment?
    a. In the pilot models, age-group and sex interaction terms were not statistically strong after adjustment. The main exception was socioeconomic status in the poor-sleep logistic model, where the screen-time-by-SES interaction was significant ($p = 0.0406$), suggesting the sleep-risk effect was steeper for

lower-SES youth. Descriptively, older youth still showed higher screen use and higher poor-sleep prevalence.

4. Which subtype of screen use (social, gaming, streaming, education) has the largest adjusted effect?
    a. It depends on the outcome. For the mental-health index, streaming and social media were the largest standardized effects, both around 0.30, with gaming slightly behind and education lower. For poor-sleep risk, gaming and streaming had the largest adjusted odds ratios, 2.513 and 2.499, but all subtype effects were adverse and fairly close in magnitude.
5. How were missing values handled, and did alternative imputation strategies change results?
    a. Milestone 2 used median imputation for low-rate missing numeric values in screen time, sleep, physical activity, anxiety, and depression. I also checked mean imputation and complete-case analysis. The results stayed very similar: the OLS screen-time coefficient was 0.213 with median imputation, 0.212 with mean imputation, and 0.223 in complete cases, while the logistic odds ratios were 2.256, 2.259, and 2.393.
6. What diagnostics were used to check model assumptions and multicollinearity?
    a. I checked variance inflation factors, heteroskedasticity, residual shape, and influence. VIF values were low, with screen_hours_total at 2.530 and the rest near or below that range, so multicollinearity was not a major issue. Breusch-

Pagan suggested mild heteroskedasticity, Jarque-Bera suggested some residual non-normality, and the maximum Cook's distance was only 0.0109, so no single case dominated the model.

7. How robust are findings if extreme high-use observations are excluded?
    a. The findings were stable. Removing the top 1% of screen-time observations above about 7.73 hours changed the OLS coefficient only from 0.213 to 0.212, and the poor-sleep odds ratio moved to 2.342. Removing observations at 9 or more hours left the OLS coefficient at 0.213 and the odds ratio at 2.247, so the core conclusion did not depend on a few extreme users.
8. What additional confounders are planned for the final model (for example family environment or baseline conditions)?
    a. The final model should add confounders where the merged real-source data support them consistently. The highest-priority additions are family environment and parental monitoring, baseline sleep or mental-health conditions, school workload, bullying or social stress exposure, bedtime device rules, caffeine use, and chronic health conditions. If those variables are not consistently available, they will be documented as residual confounding limits rather than forced into a weak harmonization.
9. How should schools apply these findings without overgeneralizing or stigmatizing students?

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a. Schools should use the findings as a support signal, not a punishment tool. The most appropriate uses are sleep education, counseling referrals, physical-activity promotion, and parent communication for high-risk patterns. The results should not be used to label individual students, justify blanket disciplinary device policies, or imply causality from screen use alone.

10. What action should the government take in addressing these findings?

    a. The government should respond with public-health guidance and better evidence infrastructure, not blunt legal screen-time limits based on one cross-sectional study. The best actions are to fund longitudinal research, improve population surveillance, support school and community sleep and mental-health programs, issue clearer parent-facing guidance, and strengthen youth-platform transparency and safety standards, especially around nighttime engagement and persuasive design.